FULL DOMAIN CANNOT BE DROPPED FROM A THEOREM ON RIGHT BREGMAN–CHEBYSHEV SETS

Abstract. Bauschke, Macklem and Wang record the following fact: if f is a Legendre function on X with $\operatorname{dom} f = X$, if $C \subseteq U = \operatorname{int}\operatorname{dom} f$ is closed, nonempty and satisfies $\operatorname{cl}\nabla f(C) \subseteq U^*$, and if C is right D_f-Chebyshev, then $\nabla f(C)$ is convex. They ask whether the hypothesis $\operatorname{dom} f = X$ is necessary. We show that it cannot be deleted. Taking f to be the negative entropy on $\mathbb{R}^2$ and $C = \{(e^t, e^{-t^2}) : t \in [1, 2]\}$, the set C is compact and right D_f-Chebyshev at every point of U, the closure hypothesis $\operatorname{cl}\nabla f(C) \subseteq U^*$ holds, and $\nabla f(C)$ is a strictly concave arc, hence nonconvex. The verification is elementary: the right Bregman projection onto C reduces to minimising a one-variable function whose second derivative is a sum of manifestly positive terms on $[1, 2]$.

The same example bears on later work of Luo, Meng, Wen and Yao. Their Theorem 3.12(2) asserts, for $U = X$, the equivalence of a variational condition (i) with the property (ii) that C is a right D_f-sun. We show by direct computation that our C is a right D_f-sun while (i) fails for an explicit point, so the requirement $U = X$ cannot be deleted from their theorem either. The obstruction is visible: the ray along which the sun property is applied leaves U after a bounded distance.

We also record, with attribution, that the remaining hypothesis $\operatorname{cl}\nabla f(C) \subseteq U^*$ is redundant, as a short consequence of the results of Luo, Meng, Wen and Yao.

1. Introduction

Let $X = \mathbb{R}^n$ with the standard inner product, and let $f : X \to \,]-\infty, +\infty]$ be proper, lower semicontinuous and convex. Write $U = \operatorname{int}\operatorname{dom} f$ and $U^* = \operatorname{int}\operatorname{dom} f^*$, where f^* is the Fenchel conjugate. Recall that f is *Legendre* if it is both essentially smooth and essentially strictly convex [1, 7]. For $x \in \operatorname{dom} f$ and $y \in U$, the *Bregman distance* is

$$D_f(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y\rangle.$$

It is not symmetric, so a set has two projection operators. Throughout, we use the *right* one, in which the second argument varies:

$$\overrightarrow{\Pi}_C(x) = \operatorname*{arg\,min}_{y\in C} D_f(x, y), \qquad x \in U.$$

Following [3] we call C *right D_f-Chebyshev* if $\overrightarrow{\Pi}_C(x)$ is a singleton for every $x \in U$. We abbreviate

$$C^* := \nabla f(C).$$

The starting point of this note is the following statement, recorded as Fact 3.2 of the survey of Bauschke, Macklem and Wang [3].

Fact 1.1 ([3, Fact 3.2])**.** Suppose that
(a) $\operatorname{dom} f = X$;
(b) $C \subseteq U$ is closed and nonempty, and $\operatorname{cl} C^* \subseteq U^*$;
(c) C is right D_f-Chebyshev.
Then C^* is convex.

The survey attaches an open problem to this fact, which we restate.

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Problem 1.2 ([3, Problem 2])**.** Is the hypothesis $\operatorname{dom} f = X$ necessary? Either exhibit a Legendre f with $\operatorname{dom} f \neq X$ and a closed nonempty right D_f-Chebyshev $C \subseteq U$ with C^* nonconvex, or show that the conclusion survives without full domain.

We settle Problem 1.2 in the first of the two directions.

Theorem 1.3. *Let $X = \mathbb{R}^2$ and let f be the negative entropy,*

$$f(x) = g(x_1) + g(x_2), \qquad g(s) = \begin{cases} s \ln s - s, & s > 0, \\ 0, & s = 0, \\ +\infty, & s < 0. \end{cases}$$

Fix real numbers a, b with $1/\sqrt{2} \le a < b$ and put

$$C = \big\{(e^t, e^{-t^2}) : t \in [a,b]\big\}.$$

Then f is Legendre with $\operatorname{dom} f = \mathbb{R}^2_+ \neq X$*, and:*

(i) *C is a nonempty compact subset of $U = \mathbb{R}^2_{++}$;*
(ii) $\operatorname{cl} C^* = C^* \subseteq U^* = \mathbb{R}^2$*, so hypothesis* (b) *of Fact 1.1 holds;*
(iii) *C is right D_f-Chebyshev, so hypothesis* (c) *holds;*
(iv) *$C^* = \{(t, -t^2) : t \in [a,b]\}$ is not convex.*

Consequently the implication of Fact 1.1 becomes false if hypothesis (a) *is deleted while* (b) *and* (c) *are retained.*

Two features of the example are worth stating at the outset, because both are easy to lose in a more elaborate construction.

First, hypothesis (b) is genuinely satisfied, not merely arranged to be vacuous. If $\operatorname{cl} C^* \subseteq U^*$ failed, the example would be deleting two hypotheses at once and would say nothing about (a) in particular. Here $U^* = \mathbb{R}^2$ and C^* is compact, so (b) holds for the strongest reason available.

Second, C is right D_f-Chebyshev at *every* point of U, including points arbitrarily close to ∂U and points far from C. Uniqueness is not a local or generic phenomenon here. It follows from a single inequality (Lemma 2.4) which holds uniformly in $x \in U$.

The verification is elementary throughout, and the proofs below use no numerical computation. The one inequality that carries the argument is

$$4t^2 - 2 \;\ge\; 0 \qquad \text{for } t \ge 1/\sqrt{2},$$

after which the required second derivative is a sum of nonnegative terms plus a strictly positive one. We draw attention to this because it is the reason the example is checkable by hand, and because it explains the constraint $a \ge 1/\sqrt{2}$, which is the only role played by the choice of interval.

Section 3 relates the example to subsequent work of Luo, Meng, Wen and Yao [5], where it turns out to have a second use, and Section 4 records what is known about the remaining hypothesis (b). Section 5 states what we do *not* claim, and what remains open.

Relation to earlier examples, and what is not claimed. The example is a variation on a template that is not ours. Bauschke, Wang, Ye and Yuan [2, Example 7.5], reproduced as Example 3.3 of the survey [3], work on the exponential side with $h(u) = e^{u_1} + e^{u_2}$ and the segment $A = \{(\lambda, 2\lambda) : 0 \le \lambda \le 1\}$, whose image under ∇h is the nonconvex curve $\{(e^\lambda, e^{2\lambda})\}$. Since the right projection for f corresponds to the left projection for f^* [5, Lemma 3.2], that curve is a nonconvex right D_f-Chebyshev set for the negative entropy, and full domain already fails there. But its gradient image is A itself, which is *convex*: the conclusion of Fact 1.1 survives, and full domain is not challenged. That is exactly why Problem 1.2 was worth asking.

Our construction changes one thing. Where [2, Example 7.5] takes the dual set to be a convex segment, we take it to be a nonconvex parabolic arc, and the work is in checking that global Chebyshevness survives the change.

	[2, Example 7.5]	Theorem 1.3
dual set A	$\{(\lambda, 2\lambda)\}$	$\{(t, -t^2)\}$
A convex?	yes	no
$C = \exp(A)$ right D_f-Chebyshev?	yes	yes
$C^* = \nabla f(C) = A$ convex?	yes	no
bears on Problem 1.2?	no	yes

Stated on the dual side, Theorem 1.3 says that the parabolic arc $\{(t, -t^2) : t \in [a, b]\}$ is a compact nonconvex set that is globally left Chebyshev for the Bregman distance of $h(u) = e^{u_1} + e^{u_2}$. This is the more transparent phrasing, and it makes clear that the obstruction lives entirely on the dual side. It is worth noting that h is not supercoercive. Since $\operatorname{dom} f = X$ is equivalent to supercoercivity of f^* for closed convex f, the dual reading of Theorem 1.3 is that a supercoercivity hypothesis cannot be deleted from the corresponding left-projection statement either.

We therefore claim neither the first nonconvex right D_f-Chebyshev set for the negative entropy — that is [2, Example 7.5] — nor the duality template, nor any new theory. Laude, Ochs and Cremers [4] also give nonconvex examples for entropy-like kernels, though their single-valuedness is local, holding on a neighbourhood under relative prox-regularity rather than at every point of U; they record Problem 1.2 as open. Themelis and Wang [6] develop Bregman operators on their natural domains, which is the setting in which a statement like Theorem 1.3 is naturally expressed.

The point of Theorem 1.3 is narrower than any of these, and is exactly what Problem 1.2 asks for: a set that is right D_f-Chebyshev *globally on* U, with *nonconvex* gradient image, *while hypothesis* (b) *holds*, so that full domain is the only hypothesis of Fact 1.1 that fails. We are not aware of an earlier example with this combination of properties.

We also emphasise what Theorem 1.3 does not say. It does not say that $\operatorname{dom} f = X$ is necessary for C^* to be convex in any individual instance; convexity of C^* certainly occurs for many f of non-full domain, [2, Example 7.5] being one. It says that $\operatorname{dom} f = X$ cannot be deleted from the *universal* statement of Fact 1.1. Nor does it identify the weakest hypothesis that could replace full domain; see Section 5.

2. The example

Throughout this section f, C, a and b are as in Theorem 1.3, and we write

$$c(t) = (e^t, e^{-t^2}), \qquad t \in [a, b],$$

so that $C = c([a, b])$. All closures are taken in $X = \mathbb{R}^2$.

Lemma 2.1. *The function f is Legendre, with*

$$\operatorname{dom} f = \mathbb{R}^2_+, \quad U = \mathbb{R}^2_{++}, \quad \nabla f(x) = (\ln x_1, \ln x_2) \quad \text{on } U,$$

and

$$f^*(u) = e^{u_1} + e^{u_2}, \quad \operatorname{dom} f^* = U^* = \mathbb{R}^2.$$

Moreover $\nabla f \colon U \to U^*$ is a bijection with inverse $u \mapsto (e^{u_1}, e^{u_2})$.

Proof. For $s > 0$ we have $g'(s) = \ln s$ and $g''(s) = 1/s > 0$, so g is strictly convex on $]0, \infty[$. Strict convexity persists at the left endpoint: for $b_0 > 0$ and $0 < \lambda < 1$,

$$g(\lambda b_0) = \lambda g(b_0) + \lambda b_0 \ln \lambda < \lambda g(b_0) + (1 - \lambda) g(0),$$

since $\ln \lambda < 0$ and $g(0) = 0$. As $\lim_{s \downarrow 0}(s \ln s - s) = 0 = g(0)$, the function g is lower semicontinuous on $\mathbb{R}$. Hence f is proper, lower semicontinuous and convex, with $\operatorname{dom} f = \mathbb{R}^2_+$ and $U = \mathbb{R}^2_{++}$. Since $(-1, 0) \notin \operatorname{dom} f$, we have $\operatorname{dom} f \neq X$.

On U the gradient is $\nabla f(x) = (\ln x_1, \ln x_2)$, and the Hessian $\operatorname{diag}(1/x_1, 1/x_2)$ is positive definite. If $x^k \in U$ converges to a boundary point of $\operatorname{dom} f$ then some coordinate tends to 0, so the corresponding component of $\nabla f(x^k)$ tends to $-\infty$ and $\|\nabla f(x^k)\| \to \infty$; thus f is essentially smooth.

No boundary point of $\operatorname{dom} f$ carries a subgradient. Indeed, suppose $x_j = 0$ and $p \in \partial f(x)$. Perturbing only the j-th coordinate to $s > 0$ in the subgradient inequality gives $s \ln s - s \geq p_j s$, hence $\ln s - 1 \geq p_j$ for every $s > 0$; taking $s = e^{p_j}$ yields $p_j - 1 \geq p_j$, a contradiction. Therefore $\operatorname{dom} \partial f = U$, and since f is strictly convex on U it is essentially strictly convex. So f is Legendre.

For the conjugate, fix $u \in \mathbb{R}$ and consider $s \mapsto us - g(s)$ on $[0, \infty[$. Its derivative on $]0, \infty[$ is $u - \ln s$ and its second derivative is $-1/s < 0$, so it is maximised uniquely at $s = e^u$, with value e^u; the boundary value at $s = 0$ is $0 < e^u$. Hence $g^*(u) = e^u$, and by separability $f^*(u) = e^{u_1} + e^{u_2}$, so $\operatorname{dom} f^* = U^* = \mathbb{R}^2$. Finally $x \mapsto (\ln x_1, \ln x_2)$ maps $\mathbb{R}^2_{++}$ bijectively onto $\mathbb{R}^2$, with the coordinatewise exponential as inverse. $\square$

Lemma 2.2. *C is nonempty and compact, $C \subseteq U$, the map c is injective, and*

$$C^* = \{(t, -t^2) : t \in [a, b]\}, \qquad \operatorname{cl} C^* = C^* \subseteq U^*.$$

Proof. The map c is continuous on the compact interval $[a, b]$, so C is compact, hence closed, and nonempty. Both coordinates of $c(t)$ are strictly positive, so $C \subseteq \mathbb{R}^2_{++} = U$. Injectivity holds because $t \mapsto e^t$ is strictly increasing. By Lemma 2.1, $\nabla f(c(t)) = (\ln e^t, \ln e^{-t^2}) = (t, -t^2)$, so C^* is the stated arc; it is a continuous image of $[a, b]$, hence compact and closed, and $U^* = \mathbb{R}^2$ contains it. $\square$

The next lemma is the computation that makes the example tractable: the right Bregman projection onto C decouples into a one-variable problem, and the part depending on x is an additive constant.

Lemma 2.3. *For every $x \in U$ and every $t \in [a, b]$,*

$$D_f(x, c(t)) = f(x) + h_x(t), \qquad \text{where} \qquad h_x(t) = e^t + e^{-t^2} - x_1 t + x_2 t^2.$$

Proof. For $x, y \in U$, expanding coordinatewise gives the familiar form

$$D_f(x, y) = \sum_{j=1}^{2} \Big[x_j \ln x_j - x_j - (y_j \ln y_j - y_j) - \ln(y_j)(x_j - y_j) \Big] = \sum_{j=1}^{2} \Big[x_j \ln \tfrac{x_j}{y_j} - x_j + y_j \Big].$$

Substituting $y = c(t) = (e^t, e^{-t^2})$ and using $\ln(x_1/e^t) = \ln x_1 - t$ and $\ln(x_2/e^{-t^2}) = \ln x_2 + t^2$,

$$D_f(x, c(t)) = \big(x_1 \ln x_1 - x_1 + x_2 \ln x_2 - x_2\big) + \big(e^t + e^{-t^2} - x_1 t + x_2 t^2\big),$$

and the first bracket is exactly $f(x)$. $\square$

Lemma 2.4. *For every $x \in U$ the function h_x is strictly convex on $[a, b]$. Indeed*

$$h_x''(t) = e^t + (4t^2 - 2)e^{-t^2} + 2x_2 \;\geq\; e^a + 2x_2 \;>\; 0 \qquad (t \in [a, b]).$$

Proof. Differentiating h_x twice gives

$$h_x'(t) = e^t - 2te^{-t^2} - x_1 + 2x_2 t, \qquad h_x''(t) = e^t + (4t^2 - 2)e^{-t^2} + 2x_2.$$

Since $t \geq a \geq 1/\sqrt{2}$ we have $4t^2 - 2 \geq 0$, so the middle term is nonnegative; and $e^t \geq e^a$ while $x_2 > 0$ because $x \in U$. The displayed bound follows, and it is strictly positive. $\square$

Remark 2.5. Only positivity is used. For the record, on $[1, 2]$ the exact infimum of $t \mapsto e^t + (4t^2 - 2)e^{-t^2}$ is attained at $t = 1$ and equals $e + 2/e = 3.45404\ldots$, so $h_x'' > e + 2/e$ there, uniformly in $x \in U$. We do not need this sharper constant, and we note that establishing it requires a genuine monotonicity argument, whereas the bound of Lemma 2.4 does not.

Lemma 2.6. *C is right D_f-Chebyshev: $\overrightarrow{\Pi}_C(x)$ is a singleton for every $x \in U$.*

Proof. Fix $x \in U$. By Lemma 2.3 the term $f(x)$ does not depend on t, so minimising $D_f(x,\cdot)$ over C is the same as minimising h_x over $[a,b]$; and by injectivity of c (Lemma 2.2) a unique minimising parameter yields a unique minimising point of C. The function h_x is continuous on the compact interval $[a,b]$, so a minimiser exists. If $s \neq t$ were both minimisers then strict convexity (Lemma 2.4) would give

$$h_x\big(\tfrac{s+t}{2}\big) < \tfrac{1}{2}\big(h_x(s)+h_x(t)\big) = \min_{[a,b]} h_x,$$

which is absurd. Hence the minimiser is unique. □

Remark 2.7. The minimiser need not be interior, and no closed form for it is needed. Since h_x' is strictly increasing by Lemma 2.4, exactly one of three cases occurs: if $h_x'(a) \geq 0$ the minimiser is $t=a$; if $h_x'(b) \leq 0$ it is $t=b$; otherwise $h_x'(a) < 0 < h_x'(b)$ and the minimiser is the unique zero of h_x' in $]a,b[$. These cases are exhaustive because $h_x'(a) < h_x'(b)$. Uniqueness in Lemma 2.6 is insensitive to which case obtains.

Lemma 2.8. *C^* is not convex.*

Proof. The points $(a,-a^2)$ and $(b,-b^2)$ lie in C^*, and their midpoint is $m = \big(\frac{a+b}{2}, -\frac{a^2+b^2}{2}\big)$. If m belonged to C^* its first coordinate would force the parameter $t=(a+b)/2$, and its second coordinate would then have to equal $-\big(\frac{a+b}{2}\big)^2$. But

$$\frac{a^2+b^2}{2} - \left(\frac{a+b}{2}\right)^2 = \frac{(a-b)^2}{4} > 0,$$

so $-\frac{a^2+b^2}{2} < -\big(\frac{a+b}{2}\big)^2$ and $m \notin C^*$. □

Proof of Theorem 1.3. Lemma 2.1 gives that f is Legendre with $\operatorname{dom} f = \mathbb{R}^2_+ \neq X$, so hypothesis (a) fails. Item (i) is Lemma 2.2, item (ii) is Lemma 2.2, item (iii) is Lemma 2.6, and item (iv) is Lemma 2.8. Since (b) and (c) hold while the conclusion of Fact 1.1 fails, hypothesis (a) cannot be deleted. □

Remark 2.9 (hypothesis audit). For the reader checking that no hypothesis has been violated silently: X is finite dimensional; f is Legendre; C is nonempty, compact, and contained in U; $\operatorname{cl} C^* \subseteq U^*$; and C is right D_f-Chebyshev on all of U. The only failure is the intended one, $\operatorname{dom} f \neq X$. If one reads Fact 1.1 under an additional standing assumption of 1-coercivity, the example still complies: $g \geq -1$ on $[0,\infty[$, so with $M = \max\{x_1,x_2\}$ we have $f(x) \geq M\ln M - M - 1$ and $\|x\|_2 \leq \sqrt{2}\,M$ for $x \in \operatorname{dom} f$, whence $f(x)/\|x\|_2 \to +\infty$; outside $\operatorname{dom} f$ the function is $+\infty$. One should not, however, read a standing assumption of supercoercivity for f^* into the setting: for a closed convex f, supercoercivity of f^* is equivalent to $\operatorname{dom} f = X$, which is exactly the hypothesis under test, so requiring it would make Problem 1.2 vacuous.

Remark 2.10. The set C itself is nonconvex, though nothing above requires this; Fact 1.1 does not assume C convex and concludes about C^*. In the variable $r = e^t$, C is the graph of $\psi(r) = e^{-(\ln r)^2}$ over $[e^a, e^b]$, and with $L = \ln r$,

$$\psi''(r) = \frac{e^{-L^2}}{r^2}\big(4L^2 + 2L - 2\big),$$

which is strictly positive for $L \geq 1/\sqrt{2}$; so for $a \geq 1/\sqrt{2}$ the function ψ is strictly convex and C is a strictly convex arc, in particular not convex as a set.

3. A consequence for the theorem of Luo, Meng, Wen and Yao

Luo, Meng, Wen and Yao [5] revisit the results surveyed in [3] with the aim of removing coercivity assumptions. Their Theorem 3.12 concerns a right D_f-proximinal set C and the statements

(i) for all $x \in U$ and $y \in C$: $y \in \overrightarrow{\Pi}_C(x)$ if and only if $\langle \nabla f(y) - \nabla f(c), y - x\rangle \le 0$ for every $c \in C$;

(ii) C is a right D_f-sun, that is, $y \in \overrightarrow{\Pi}_C(x)$ implies $y \in \overrightarrow{\Pi}_C(\lambda x + (1-\lambda)y)$ for every $\lambda \ge 0$ with $\lambda x + (1-\lambda)y \in U$;

(iv) $\nabla f(C)$ is convex.

They prove that (i) $\Rightarrow$ (ii) with no domain hypothesis; that (i) $\Leftrightarrow$ (ii) *if* $U = X$; and that (i) $\Leftrightarrow$ (iv) if $\nabla f(U) = U^*$ and f^* is Gâteaux differentiable and strictly convex on U^*. Their Theorem 3.13 states that if f is totally convex at every point of U, if $\nabla f(U) = U^*$, and if f^* is locally uniformly totally convex at every point of U^*, then every boundedly compact right D_f-Chebyshev set is a right D_f-sun.

Our example satisfies (ii) but not (i), and does not satisfy $U = X$. Both halves can be checked directly, without invoking either of their theorems, and we do so: the point is to exhibit a configuration their Theorem 3.12(2) would forbid, so it is better not to rely on their machinery to produce it.

Lemma 3.1. *Let f and C be as in Theorem 1.3. Then C is a right D_f-sun.*

Proof. Write $y = c(s) \in \overrightarrow{\Pi}_C(x)$ for some $x \in U$ and $s \in [a, b]$, and let $\lambda \ge 0$ satisfy $z_\lambda := \lambda x + (1-\lambda)y \in U$. By Lemma 2.3, the projection of any $w \in U$ is governed by h_w, and by Lemma 2.4 each h_w is strictly convex on $[a, b]$, so h'_w is strictly increasing.

The essential observation is that

$$h'_w(t) = e^t - 2te^{-t^2} - w_1 + 2w_2 t$$

is an *affine* function of w. Moreover $h'_y(s) = 0$: substituting $w = y = c(s) = (e^s, e^{-s^2})$ gives

$$h'_y(s) = e^s - 2se^{-s^2} - e^s + 2e^{-s^2}s = 0.$$

Hence, by affinity,

$$h'_{z_\lambda}(s) = \lambda h'_x(s) + (1-\lambda)h'_y(s) = \lambda\, h'_x(s).$$

Now use the three cases of Remark 2.7 for x. If $s \in \,]a, b[$ then $h'_x(s) = 0$, so $h'_{z_\lambda}(s) = 0$ and s is the unique minimiser of h_{z_λ}. If $s = a$ then $h'_x(a) \ge 0$, so $h'_{z_\lambda}(a) = \lambda h'_x(a) \ge 0$ since $\lambda \ge 0$, and as h'_{z_λ} is increasing the unique minimiser is a. If $s = b$ then $h'_x(b) \le 0$, so $h'_{z_\lambda}(b) \le 0$ and the unique minimiser is b. In all three cases $\overrightarrow{\Pi}_C(z_\lambda) = \{c(s)\} = \{y\}$. □

Lemma 3.2. *Let f and C be as in Theorem 1.3, put $s = (a+b)/2$ and*

$$x = \big(e^s - se^{-s^2},\ \tfrac{1}{2}e^{-s^2}\big).$$

Then $x \in U$ and $y := c(s) \in \overrightarrow{\Pi}_C(x)$, but for every $t \in [a, b]$ with $t \ne s$,

$$\langle \nabla f(y) - \nabla f(c(t)), y - x\rangle = (s-t)^2\, \tfrac{1}{2}e^{-s^2} > 0.$$

Hence statement (i) *fails.*

Proof. Both coordinates of x are positive, since $e^s > se^{-s^2}$ for $s \ge 1/\sqrt{2}$, so $x \in U$. With $x_2 = \frac{1}{2}e^{-s^2}$ we get $2s\big(e^{-s^2} - x_2\big) = se^{-s^2} = e^s - x_1$, which rearranges to $h'_x(s) = e^s - 2se^{-s^2} - x_1 + 2x_2 s = 0$. As $s \in \,]a, b[$ and h_x is strictly convex, s is the unique minimiser, so $y = c(s)$ is the right projection of x.

By Lemma 2.1, $\nabla f(c(r)) = (r, -r^2)$, so

$$\nabla f(y) - \nabla f(c(t)) = (s-t,\ t^2 - s^2) = (s-t)\,(1,\ -(s+t)),$$

while $y - x = \big(e^s - x_1,\ e^{-s^2} - x_2\big)$. Therefore

$$\langle \nabla f(y) - \nabla f(c(t)), y - x\rangle = (s-t)\Big[(e^s - x_1) - (s+t)\big(e^{-s^2} - x_2\big)\Big].$$

Substituting $e^s - x_1 = 2s(e^{-s^2} - x_2)$ from the stationarity relation above, the bracket becomes $(2s - s - t)(e^{-s^2} - x_2) = (s - t)(e^{-s^2} - x_2)$, so the whole expression equals $(s-t)^2(e^{-s^2} - x_2) = (s-t)^2 \cdot \frac{1}{2}e^{-s^2}$, which is strictly positive for $t \neq s$. Since $a < b$ such a t exists in $[a, b]$. Statement (i) requires this quantity to be ≤ 0 for every $c \in C$, so (i) fails. □

Corollary 3.3. *Let f and C be as in Theorem 1.3. Then* (ii) *holds,* (i) *and* (iv) *fail, and $U \neq X$. Consequently the hypothesis $U = X$ in* [5, Theorem 3.12(2)] *cannot be deleted without replacement: the implication* (ii) $\Rightarrow$ (i) *is false without it.*

Proof. Statement (ii) is Lemma 3.1, statement (i) fails by Lemma 3.2, and (iv) fails by Lemma 2.8. Finally $U = \mathbb{R}^2_{++} \neq \mathbb{R}^2 = X$. □

We state the conclusion as non-omissibility rather than necessity. The example shows that $U = X$ cannot simply be struck from Theorem 3.12(2); it does not show that $U = X$ is the weakest hypothesis that would serve there, and some weaker ray-completeness condition may well suffice.

Remark 3.4 (where the obstruction lies). The example locates the failure precisely. The proof of (ii) $\Rightarrow$ (i) in [5, Theorem 3.12(2)] applies the sun property along z_λ for arbitrarily large λ, divides by λ, and lets $\lambda \to \infty$. For the witness of Lemma 3.2 that limit is unavailable: with $\eta = e^{-s^2}$ we have $(z_\lambda)_2 = \lambda\frac{\eta}{2} + (1-\lambda)\eta = \eta\big(1 - \frac{\lambda}{2}\big)$, so $z_\lambda \in U$ exactly for $0 \leq \lambda < 2$. The ray leaves U after a bounded distance, and it is this bounded reach, not the sun property itself, that $U = X$ was supplying.

Remark 3.5. No result of [5] is contradicted. Their Theorem 3.12(1) gives (i) $\Rightarrow$ (ii) unconditionally, which is consistent with (ii) holding and (i) failing. Their Theorem 3.12(3) gives (i) $\Leftrightarrow$ (iv) under hypotheses our example does satisfy, and indeed (i) and (iv) fail together here, as Lemmas 2.8 and 3.2 confirm independently. Their Theorem 3.13 also applies to our example, by a routine verification that negative entropy is totally convex on U and that f^* is locally uniformly totally convex on U^*; it yields Lemma 3.1 again, and we gave the direct proof instead because it is shorter and does not depend on that verification. Their Corollary 3.14, which does conclude convexity of $\nabla f(C)$ from bounded compactness and right Chebyshevness, assumes total convexity at every point of X and $\nabla f(X) = U^*$; both presuppose $U = X$ and therefore do not apply here.

Remark 3.6. It is instructive that the same example does duty twice. Both [3, Fact 3.2] and [5, Theorem 3.12(2)] reach convexity of $\nabla f(C)$ from a Chebyshev-type hypothesis, and both require full domain at the corresponding step, one by assuming $\operatorname{dom} f = X$ outright and the other by requiring $U = X$. Theorem 1.3 and Corollary 3.3 together show the requirement is not an artefact of either proof technique.

4. The remaining hypothesis

For completeness we record the status of hypothesis (b) of Fact 1.1, which is not ours to claim.

Proposition 4.1 (essentially [5, Theorems 3.12 and 3.13])**.** *Let f be Legendre on $X = \mathbb{R}^n$ with $\operatorname{dom} f = X$, and let $C \subseteq X$ be nonempty and closed. If C is right D_f-Chebyshev, then C^* is convex. In particular hypothesis* (b) *of Fact 1.1 is redundant.*

Proof. Since $\operatorname{dom} f = X$ we have $U = X$, and C, being closed in $\mathbb{R}^n$, is boundedly compact. By [7, Theorem 26.5], ∇f is a bijection of $U = X$ onto U^* with inverse ∇f^*, and f^* is Legendre; in particular $\nabla f(U) = U^*$, and f^* is differentiable and strictly convex on U^*.

The total-convexity hypotheses of [5, Theorem 3.13] are automatic here. For $y \in X$ and $t > 0$ the set $\{z \in \operatorname{dom} f = X : \|z - y\| = t\}$ is compact, $D_f(\cdot, y)$ is continuous on it, and $D_f(z, y) > 0$ for $z \neq y$ by strict convexity, so $\nu_f(y, t) > 0$. For f^*, fix $u_0 \in U^*$ and $t > 0$ and choose $\rho > 0$ with $\overline{B}(u_0, \rho) \subseteq U^*$; the *joint* set $K = \{(u, v) : \|u - u_0\| \leq \rho,\ \|v - u\| = t\}$ is compact and contains no pair with $v = u$, so $m := \min_K D_{f^*} > 0$ and $\nu_{f^*}(u,t) \geq m$ for every $u \in \overline{B}(u_0, \rho)$, whence $\nu^{\mathrm{loc}}_{f^*}(u_0, t) \geq m > 0$. (A bound for each fixed u would not suffice, since such bounds could degenerate as $u \to u_0$.)

So [5, Theorem 3.13] gives that C is a right D_f-sun, which is (ii); [5, Theorem 3.12(2)], applicable since $U = X$, yields (i); and [5, Theorem 3.12(3)] makes (i) equivalent to convexity of $\nabla f(C)$. □

Together with Theorem 1.3, this completes the account of Fact 1.1: hypothesis (a) is essential, and hypothesis (b) is not needed.

5. What remains open

Theorem 1.3 answers Problem 1.2 as posed, but leaves the more interesting half of the question untouched.

(1) *What replaces full domain?* Our example shows $\operatorname{dom} f = X$ cannot simply be deleted. It does not identify the weakest hypothesis under which the conclusion survives. A natural guess is a condition on how C^* approaches ∂U^*, since by the duality of [5, Lemma 3.2] the right projection onto C for f is the left projection onto C^* for f^*, and the obstruction in our example lives entirely on the dual side.

(2) *Problem 4 of* [3] asks for a characterisation of the right D_f-Chebyshev sets for the negative entropy. A solution would subsume Theorem 1.3, which exhibits one such set and computes its gradient image. We regard this as the natural next question, and note that our example is a member of a one-parameter family (any $1/\sqrt{2} \leq a < b$), which suggests such sets are not scarce.

(3) *Dimension and geometry.* Our example is planar and our curve is an arc. We do not know whether the failure can be made worse, for instance with C^* having nonempty interior in its convex hull, nor what the analogous picture is in $\mathbb{R}^n$ for $n \geq 3$.

Verification

The claims of Theorem 1.3 are elementary and were checked by hand, but they were also checked numerically before and after the proofs were written. A harness verifies, in directed-rounding interval arithmetic and to 40 digits: that D_f agrees with its definition and with the generalised Kullback–Leibler form; that the reduction of Lemma 2.3 holds with the constant equal to $f(x)$; that h_x'' is bounded below uniformly on $[1,2]$; that the minimiser of h_x is unique for a spread of adversarial $x \in U$ including near-boundary and far-field points; and the nonconvexity witness of Lemma 2.8. A second harness tests Lemma 3.1 directly against the definition of a sun, over 1,038 pairs (x, λ) with λ ranging beyond x to the boundary of U and including points constructed to project to the interior of the arc; the projection parameter is invariant to 4×10^{-40}. It also confirms the witness of Lemma 3.2, both the closed form $(s-t)^2\big(e^{-s^2} - x_2\big)$ for the inner product and its positivity. These checks are finite and do not certify the universal quantifiers, which are carried by the proofs above.

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